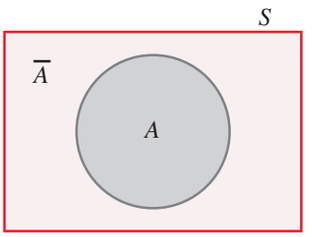
### Theorem 12.4

If $\overline{A}$ is the complementary event of $A$, then

$$\boxed{P(\overline{A}) = 1 - P(A)}$$




**Proof**

Let $S$ be a sample space, we have

$$A \cup \overline{A} = S$$

$$P(A \cup \overline{A}) = P(S)$$

$$P(A) + P(\overline{A}) = P(S) \text{ (since } A \text{ and } \overline{A} \text{ are mutually exclusive)}$$

$$= 1$$

$$P(\overline{A}) = 1 - P(A) \text{ or } P(A) = 1 - P(\overline{A})$$

**Example 12.13**

Nine coins are tossed once, find the probability to get at least two heads.

**Solution**

Let $S$ be the sample space and $A$ be the event of getting at least two heads.

Therefore, the event $\overline{A}$ denotes, getting at most one head.

$$n(S) = 2^9 = 512, \quad n(\overline{A}) = 9C_0 + 9C_1 = 1 + 9 = 10$$

$$P(\overline{A}) = \frac{10}{512} = \frac{5}{256}$$

$$P(A) = 1 - P(\overline{A}) = 1 - \frac{5}{256} = \frac{251}{256}$$

It is easier to calculate $\sum_{r=0}^{1} 9C_r$ than $\sum_{r=2}^{9} 9C_r$

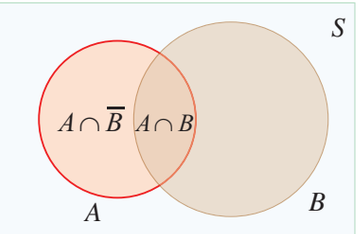
### Theorem 12.5

If $A$ and $B$ are any two events and $\overline{B}$ is the complementary events of $B$, then

$$\boxed{P(A \cap \overline{B}) = P(A) - P(A \cap B)}$$

**Proof**

Clearly from the figure,




$$(A \cap \overline{B}) \cup (A \cap B) = A$$

$$P\left[(A \cap \overline{B}) \cup (A \cap B)\right] = P(A)$$

$$P(A \cap \overline{B}) + P(A \cap B) = P(A)$$

(since $(A \cap \overline{B})$ and $(A \cap B)$ are mutually exclusive)

$$P(A \cap \overline{B}) = P(A) - P(A \cap B)$$



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